Chandrachud Pati

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Research Interests

Reliable quantum computing: quantum compilers and compiler diagnostics; measurement budgets and QPU-time estimation; quantum–HPC runtime systems and workflows; noise- and calibration-aware execution on quantum hardware.

Education

National Institute of Technology Rourkela 2019–2024
Integrated M.Sc. in Mathematics *Rourkela, India*

- **CGPA:** 7.72/10; first three years 8.39/10.
- **Selected coursework:** Probability and Statistics, Mathematical Methods, Numerical Analysis, Linear Programming, Statistics Laboratory, Basic Programming, and Computing Laboratory.

Publications

[1] **Chandrachud Pati** and Yogesh Simmhan. "Per-Phase Fidelity Attribution for Quantum Compilers using HBR Decomposition." *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026. arXiv:2605.07876.

[2] **Chandrachud Pati**, Abdur Rahman Hatim, and Yogesh Simmhan. "Representation-Aware Portability of Quantum Circuits Across SDK Boundaries." *33rd IEEE International Conference on High Performance Computing, Data & Analytics (HiPC)*, 2026.

[3] **Chandrachud Pati** and Yogesh Simmhan. "SemQBench: Benchmarking Decomposition-Aware Runtime Policies for Quantum-HPC Search." *2nd International Workshop SFWM, co-located with SC*, 2026. arXiv:2610.22886.

[4] **Chandrachud Pati** and Yogesh Simmhan. "Schedulable but Infeasible: Pre-dispatch Checks for Quantum HPC Workflows." Poster, *IEEE HiPC*, 2026. Interactive demo.

[5] Sakshi Chhabra*, **Chandrachud Pati***, and Yogesh Simmhan. "Towards Forecasting Calibration Windows for Qubit Placement on IBM Quantum Systems." Poster, *IEEE HiPC*, 2026. (*Equal contribution.)

[6] Aryan Singh Sisodiya, **Chandrachud Pati**, and Yogesh Simmhan. "Pre-Execution QPU Time Estimation for Quantum Jobs in HPC Systems." Poster, *IEEE HiPC*, 2026.

Manuscripts Under Review

[7] **Chandrachud Pati** et al. Title withheld for double-blind review. *IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 2027.

[8] **Chandrachud Pati** et al. Title withheld for double-blind review. *IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 2027.

[9] **Chandrachud Pati** et al. Title withheld for double-blind review. *ACM SIGMETRICS*, 2027.

Research Experience

Indian Institute of Science (IISc) Feb 2026–Present
Pre-doctoral Researcher / Project Associate-I, DREAM Lab *Bengaluru, India*
Advisor: Prof. Yogesh Simmhan, Department of Computational and Data Sciences.

- **Quantum compilers.** Conceived and led HBR decomposition, which attributes fidelity loss to individual compiler phases (high-level decomposition, basis translation, and routing) instead of treating the compiler as a black box; instrumented Qiskit, PennyLane, and TKET and confirmed the findings with noisy simulation and IBM Fez runs [1].
- Measured how a circuit's SDK construction-and-export path changes its compiled cost across six SDKs and nine workloads at 8–80 qubits; at 80 qubits, two SDKs' Grover exports need 10.3× and 15.6× as many two-qubit gates as the other four [2].
- **Measurement and resource estimation.** Built a measurement service for variational algorithms that reuses variance bounds from earlier requests while guaranteeing the requested precision; on 40 H_2 and LiH optimizations it used 35% fewer shots and cost 19% less than the strongest guaranteed pilot baseline [9].
- Built QFO, which checks before dispatch whether a quantum–classical plan can meet an application's error target [4]; with postgraduate students, worked on pre-execution QPU-time estimation using graph neural networks [6] and on forecasting calibration updates to decide when to place jobs on IBM Heron processors [5].
- **Quantum–HPC systems.** Under India's National Quantum Mission, building middleware that optimizes portable quantum–HPC workflows [7], benchmarking decomposition-aware runtime policies on IQM hardware [3], and recovering application decisions when retried workflow tasks return different valid results [8].

National Institute of Technology Rourkela Sep 2023–Jan 2024
Research Project, Hybrid Quantum–Classical Machine Learning *Rourkela, India*
Advisor: Prof. Snehashish Chakraverty.

- Designed a hybrid neural network with two-qubit parameterized circuits, implemented in Qiskit and PennyLane as trainable Keras layers, for highly imbalanced credit-card fraud detection.
- Improved AUC by about two percentage points over a matched classical baseline, and found that transpilation fidelity changed model accuracy, which motivated the later compiler work.

Industry Experience

American Express Aug 2024–Jan 2026
Engineer II, Next Generation Data Management *Gurugram, India*

- Built an LLM-based data-architecture documentation system that reduced manual documentation effort by approximately 60% and contributed to an early promotion to Engineer II.
- Developed Python REST APIs for infrastructure provisioning and worked with enterprise-scale distributed data systems.

IDFC FIRST Bank Feb 2024–Jun 2024
Data Science Intern, New Age Special Projects *Mumbai, India*

- Fine-tuned BERT and open-source language models for real-time transaction classification, reaching approximately 99% accuracy on an internal evaluation set; built PySpark and AWS pipelines for large-scale transaction records.

Tally Solutions May 2023–Jul 2023
Engineering Intern, CTO's Office *Bengaluru, India*

- Designed Go chaincode and REST APIs for a Hyperledger Fabric platform supporting MSME credit workflows; containerized peers, orderers, certificate authorities, and application services.

Leadership and Service

Technical Society Secretary, NIT Rourkela; elected student representative, ran technical events at Innovision (12,000+ attendees). 2023–2024

Coordinator, Career Development Centre, NIT Rourkela; trained students in software and analytics. 2023–2024

Chief Coordinator, editorial board of the campus student media body; led the rebuild of its website (350,000 visits). 2021–2022

Core Organizer, HackNITR 3.0, one of India's largest student-run hackathons (5,000+ participants). 2021

Honors and Awards

2026 **ACM SIGMOD/PODS India Fellowship**, recipient.
2025 **American Express Early Careers Hackathon**, winner; synthetic-data generation with GANs.
2023 **IDFC FIRST Bank FAME 2.0**, runner-up; EMI default prediction and collections modelling.
2023 **Flipkart GRID 5.0**, semi-finalist; quantum computing for diabetic-retinopathy detection.
2023 **Unstop UG Leaders**, All-India Rank 4.
2022 **Tally CodeBrewers**, national runner-up; systems programming for file tagging.

Technical Skills

Quantum: Qiskit, PennyLane, TKET, Cirq, Braket; transpilation, noisy simulation, backend-aware benchmarking, resource estimation, quantum–HPC workflows.
Systems and HPC: Python, Slurm, Parsl, PySpark, SQL, AWS, Docker, Kubernetes, Git, Linux; distributed workflows and enterprise data platforms.
Machine learning: PyTorch, TensorFlow/Keras, graph neural networks, LLM fine-tuning, retrieval-augmented generation, GANs.